

Sums of independent random variables with unbounded variance

OPG-36887

L1 - TRACTABLE

PARTIALLY SOLVED

Computer Science

ORIGINAL PROBLEM STATEMENT

Conjecture If $X_1, \dots, X_n \geq 0$ are independent random variables with $\mathbb{E}[X_i] \leq \mu$, then

$$\Pr\left(\sum X_i - \mathbb{E}\left[\sum X_i\right] < \delta\mu\right) \geq \min\left((1 + \delta)^{-1}\delta, e^{-1}\right).$$